



SAINT PIERRE
LEARNING RESOURCES

BIOLOGY

General Biology

Cell Membrane Transport

College • General Biology I

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Teacher Guide

Level: College • General Biology I

Standard / objective: Course objective — Predict passive and active transport outcomes using concentration gradients, membrane properties, and energy requirements.

Source / course note: Course-level objective; no accreditation or universal equivalency claimed.

Before this lesson: Identify a membrane, solute, solvent, and concentration.

Model and discuss: Assume water crosses but the solute cannot. With 2% solute inside and 8% outside, net water movement is out. State this permeability assumption before predicting.

Watch for: Tonicity depends on nonpenetrating solutes; a percentage comparison alone is incomplete without membrane assumptions.

Suggested sequence: 5-minute launch, 10-minute model, 20-30 minutes of practice, then a 5-minute exit ticket. Use separate paper for extended writing.

Learning supports

AIG: Explain how the prediction changes if the solute crosses the membrane freely.

504: Offer chunked tasks, reduced copying, extra time, and breaks as specified in the learner's plan.

EC: Draw a labeled membrane and arrows; count particles using the same volume on each side.

ESL: Rehearse: Net water moves ___ because the nonpenetrating solute concentration is ___.

Adapt these suggestions to the current learning plan. Print on US Letter paper at actual size; keep the key separate. One teacher may use the resource with their own students; no public posting or resale.

Student Directions & Practice

Use concentration scenarios to compare diffusion, osmosis, facilitated diffusion, and active transport.

Directions: Complete every item. Show work or cite evidence. For open responses, answer in complete sentences.

- 1.** Define concentration gradient.
- 2.** Does simple diffusion require ATP?
- 3.** Water can cross the membrane, but the solute cannot. Predict net water movement: cell 2% solute, solution 8% solute.
- 4.** Name transport through a channel protein down a gradient.
- 5.** Why can ions not freely cross the lipid bilayer?
- 6.** Which process moves solute against its gradient?
- 7.** Predict a red blood cell in a strongly hypotonic solution.
- 8.** Design a control for a dialysis-tubing investigation.



Assessment / Exit Ticket

Work independently. Show your reasoning. Use extra paper if needed.

1. A cell has 3% nonpenetrating solute; the solution has 1%. Water can cross. Predict net water movement.

2. A carrier moves solute against its gradient using energy. Identify the transport type and explain.

Reflection: What helped you decide? What do you want to practice next?

Score each task: 2 = accurate with clear reasoning; 1 = partly correct or missing reasoning; 0 = incorrect or no evidence. The reflection is not scored.

Answer Key & Teaching Guidance

1. A difference in concentration across space or a membrane.
2. No.
3. Water moves out of the cell toward the higher solute concentration.
4. Facilitated diffusion.
5. Their charge interacts poorly with the hydrophobic membrane interior.
6. Active transport; it requires energy.
7. Water enters; the cell swells and may lyse.
8. An identical setup with no concentration difference or without the tested solute.

Exit ticket guidance

1. Water moves into the cell; the external solution is hypotonic under the stated assumptions.
2. Active transport. Movement against the gradient requires an energy source.

Next teaching move: Use the worked model again if the concept is unclear. If the answer is correct but reasoning is incomplete, ask for an explanation. Offer the extension once both tasks are secure. Scores guide instruction; they are not standardized mastery cutoffs.